

实验四 中规模组合逻辑

第一部分 预习报告

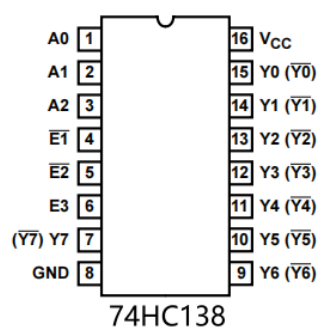
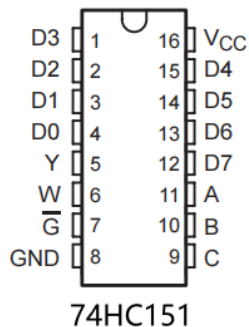
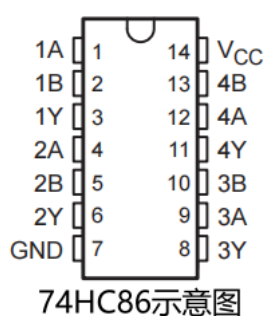
必做实验

一、设计一个 3 位二进制原码转补码电路，用三种方案实现（第 9 周检查预搭，课内验收）

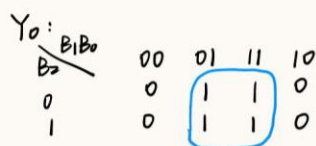
注意：不考虑符号位，默认为 0 和负数，异或逻辑可以直接选用 7486

真值表

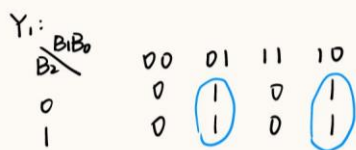
B2	B1	B0	Y2	Y1	Y0
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	0	0
1	0	1	0	1	1
1	1	0	0	1	0
1	1	1	0	0	1



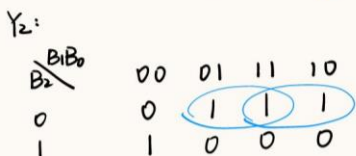
卡诺图:



$$Y_0 = B_0$$

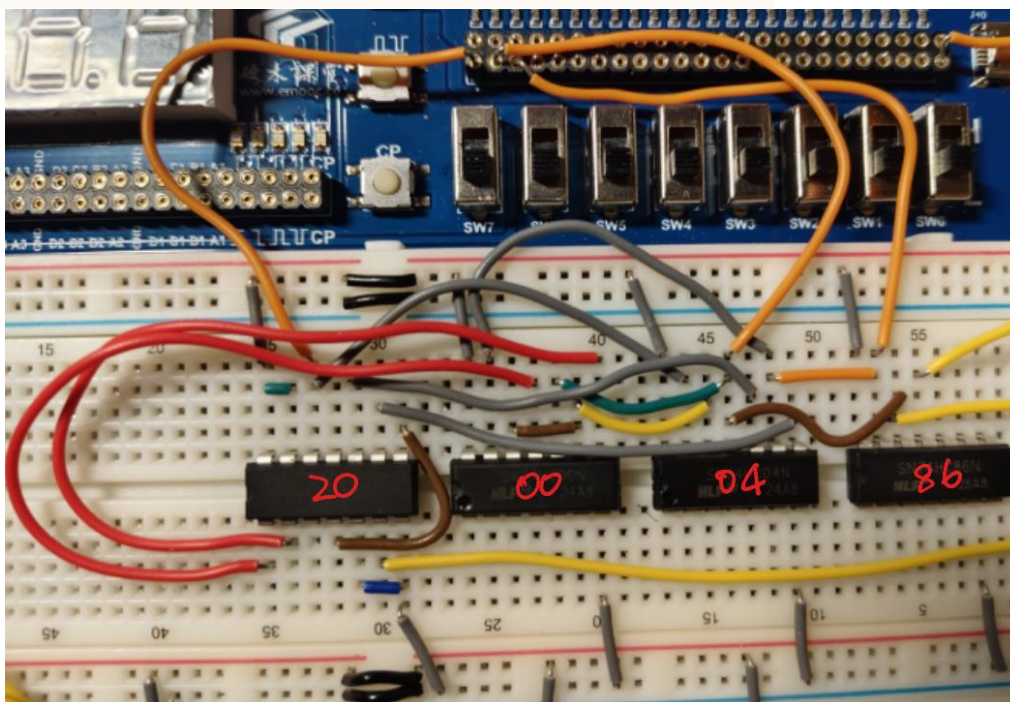
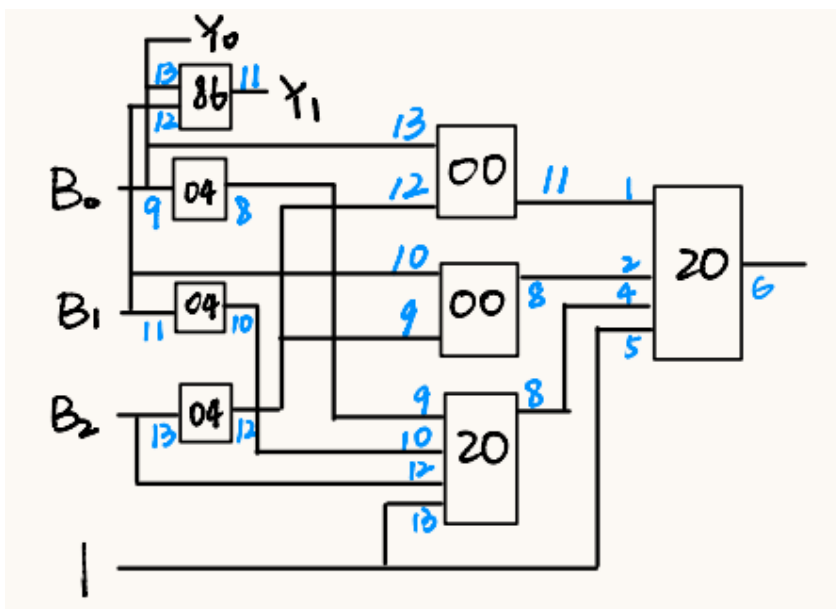


$$Y_1 = \bar{B}_1 B_0 + B_1 \bar{B}_0 = B_1 \oplus B_0$$

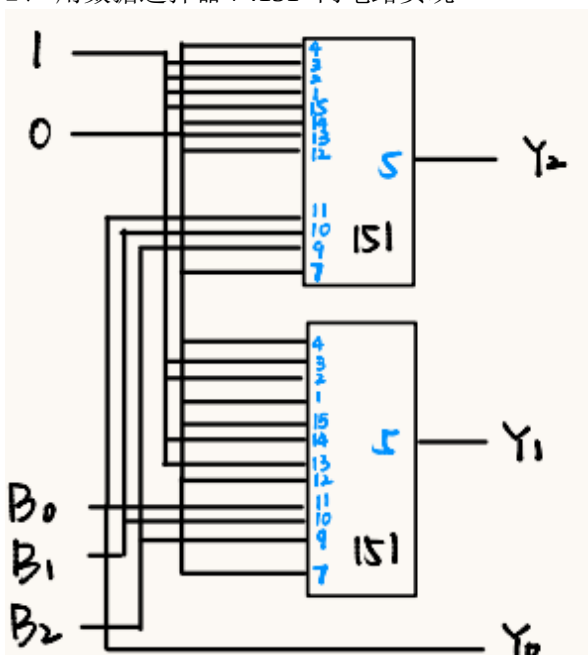


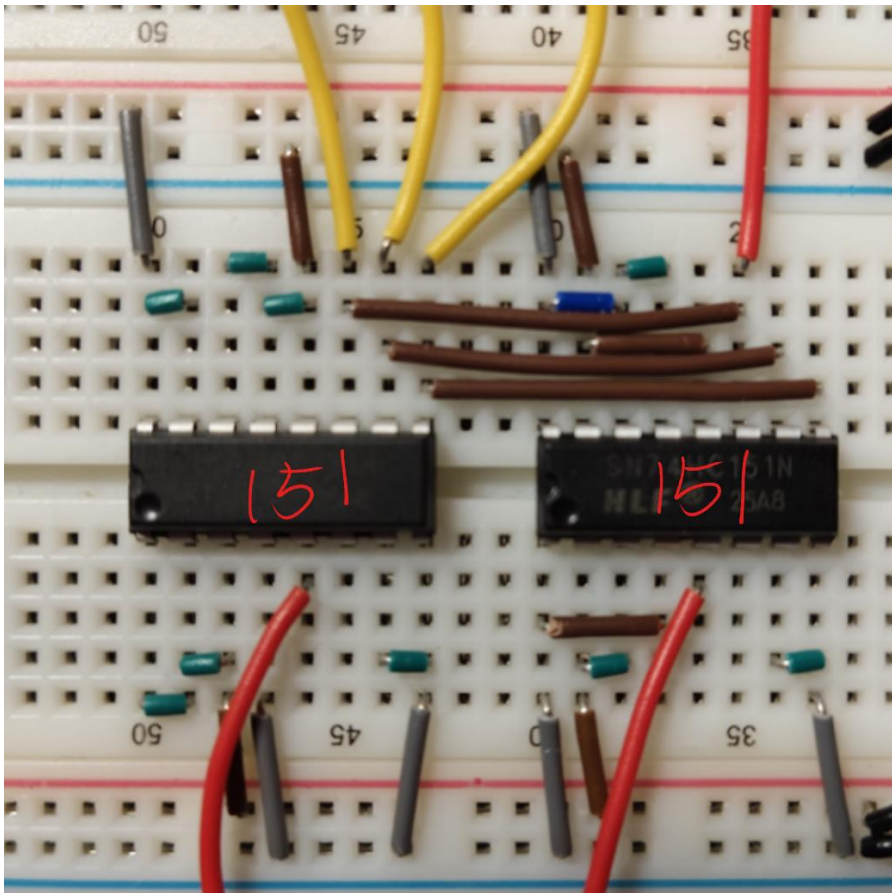
$$Y_2 = B_2 \bar{B}_1 \bar{B}_0 + \bar{B}_2 B_0 + \bar{B}_2 B_1 = \overline{B_2 \bar{B}_1 \bar{B}_0 \cdot \bar{B}_2 B_1 \cdot \bar{B}_2 B_0}$$

1、全部用门电路实现，提示：异或逻辑可以直接选用 7486



2、用数据选择器 74151+门电路实现





3、用三八译码器 74138+门电路实现

